

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Camera
Manufacturer: Avonic
Firmware Version: N/A
Model(s): CM60-IP

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.05.0000-b007	2.01.0002-b005	1.2.3

Version History

Module Version	Date	Notes
1_0_0_0	4/13/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example: InterfaceName.connectionCounter = 5
- CameraAddress variable must be set to the Address on the camera. Default value is '1'. Camera Address ranges from '1' – '7'. Example: InterfaceName.CameraAddress = '1'

Control Commands, States, and Qualifiers:

Aperture ¹	'Reset'	'Up'	'Down'
AutoFocus	'On'	'Off'	
AutomaticExposure ¹	'Full Auto' 'Iris Priority'	'Manual' 'Bright'	'Shutter Priority'
BacklightMode ¹	'On'	'Off'	
Focus	'Stop'	'Far'	'Near'
['Speed']	'0' – '7'		
Iris ¹	'Reset'	'Up'	'Down'
PanTilt	'Up' 'Right' 'Down Left' 'Home'	'Down' 'Up Left' 'Down Right' 'Reset' ¹	'Left' 'Up Right' 'Stop'
['Pan Speed']	'1' – '24'		
['Tilt Speed']	'1' – '20'		
Power ¹	'On'	'Off'	
Preset	'Reset'	'Set'	'Recall'
['Number'] ²	0 to 254 in steps of 1		
Zoom	'Stop'	'Tele'	'Wide'
['Speed']	'0' – '7'		

¹ Applies for serial control only.

² For serial control, the max number is 127.

Status Available:

AutoFocus ¹	'On'	'Off'	
AutomaticExposure ¹	'Full Auto'	'Manual'	'Shutter Priority'
	'Iris Priority'	'Bright'	
BacklightMode ¹	'On'	'Off'	
ConnectionStatus	'Connected'	'Disconnected'	
Power ¹	'On'	'Off'	'Internal Power Circuit Error'
RequiredPolling ²	None		

¹ Applies for serial control only.

² Applies for ethernet control only. Supports Update only and is used only to trigger ConnectionStatus.

Cable and Adapter Requirements:

Captive Screw to DIN-8 VISCA Serial Cable

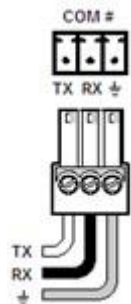
Notes for the Device:

Serial communication:

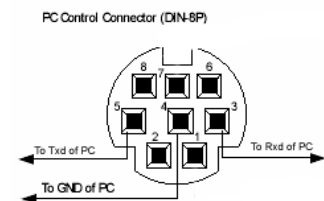
Port Type: RS-232
Baud Rate: 9600
Data Bits: 8

Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	5	RxD
RxD	←	3	TxD
GND	→	4	GND



General Notes:

Network communication:

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

Port Type:	Ethernet
Default Port:	5678
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	No

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Appendix A. Set Commands (Serial)

Aperture Down	\x81\x01\x04\x02\x03\xff
Aperture Reset	\x81\x01\x04\x02\x00\xff
Aperture Up	\x81\x01\x04\x02\x02\xff
Auto Focus Off	\x81\x01\x048\x03\xff
Auto Focus On	\x81\x01\x048\x02\xff
Automatic Exposure Bright	\x81\x01\x049\x0D\xff
Automatic Exposure Full Auto	\x81\x01\x049\x00\xff
Automatic Exposure Iris Priority	\x81\x01\x049\x0B\xff
Automatic Exposure Manual	\x81\x01\x049\x03\xff
Automatic Exposure Shutter Priority	\x81\x01\x049\x0A\xff
Backlight Mode Off	\x81\x01\x043\x03\xff
Backlight Mode On	\x81\x01\x043\x02\xff
Focus Far Speed 0	\x81\x01\x04\x08\xff
Focus Far Speed 7	\x81\x01\x04\x08\xff
Focus Near Speed 0	\x81\x01\x04\x080\xff
Focus Near Speed 7	\x81\x01\x04\x087\xff
Focus Stop Speed 0	\x81\x01\x04\x08\x00\xff
Focus Stop Speed 7	\x81\x01\x04\x08\x00\xff
Iris Down	\x81\x01\x04\x0B\x03\xff
Iris Reset	\x81\x01\x04\x0B\x00\xff
Iris Up	\x81\x01\x04\x0B\x02\xff
Pan-Tilt Down Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\xff
Pan-Tilt Down Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\xff
Pan-Tilt Down Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\xff
Pan-Tilt Down Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\xff
Pan-Tilt Down Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x02\xff
Pan-Tilt Down Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x02\xff
Pan-Tilt Down Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x02\xff
Pan-Tilt Down Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x02\xff
Pan-Tilt Down Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x02\xff
Pan-Tilt Down Right Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\x02\xff
Pan-Tilt Down Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x02\xff
Pan-Tilt Down Right Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\x02\xff
Pan-Tilt Home Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x04\xff
Pan-Tilt Home Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x04\xff
Pan-Tilt Home Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x04\xff
Pan-Tilt Home Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x04\xff
Pan-Tilt Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\xff
Pan-Tilt Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x01\x03\xff
Pan-Tilt Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x03\xff
Pan-Tilt Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x01\x03\xff
Pan-Tilt Reset Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x05\xff
Pan-Tilt Reset Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x05\xff
Pan-Tilt Reset Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x05\xff
Pan-Tilt Reset Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x05\xff

Pan-Tilt Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x02\x03\xFF
Pan-Tilt Right Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\x03\xFF
Pan-Tilt Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x03\xFF
Pan-Tilt Right Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\x03\xFF
Pan-Tilt Stop Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x03\xFF
Pan-Tilt Stop Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x03\xFF
Pan-Tilt Stop Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x03\xFF
Pan-Tilt Stop Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x03\xFF
Pan-Tilt Up Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x01\x01\xFF
Pan-Tilt Up Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x01\x01\xFF
Pan-Tilt Up Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x01\xFF
Pan-Tilt Up Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x01\x01\xFF
Pan-Tilt Up Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x01\xFF
Pan-Tilt Up Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x01\xFF
Pan-Tilt Up Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x01\xFF
Pan-Tilt Up Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x01\xFF
Pan-Tilt Up Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x01\xFF
Pan-Tilt Up Right Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\x01\xFF
Pan-Tilt Up Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x01\xFF
Pan-Tilt Up Right Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\x01\xFF
Power Off	\x81\x01\x04\x00\x03\xFF
Power On	\x81\x01\x04\x00\x02\xFF
Preset Recall Number 0	\x81\x01\x04?\x02\x00\xFF
Preset Recall Number 127	\x81\x01\x04?\x02\x7F\xFF
Preset Reset Number 0	\x81\x01\x04?\x00\x00\xFF
Preset Reset Number 127	\x81\x01\x04?\x00\x7F\xFF
Preset Set Number 0	\x81\x01\x04?\x01\x00\xFF
Preset Set Number 127	\x81\x01\x04?\x01\x7F\xFF
Zoom Stop Speed 0	\x81\x01\x04\x07\x00\xFF
Zoom Stop Speed 7	\x81\x01\x04\x07\x00\xFF
Zoom Tele Speed 0	\x81\x01\x04\x07 \xFF
Zoom Tele Speed 7	\x81\x01\x04\x07\xFF
Zoom Wide Speed 0	\x81\x01\x04\x070\xFF
Zoom Wide Speed 7	\x81\x01\x04\x077\xFF

Appendix B. Query Commands (Serial)

Automatic Exposure	\x81\x09\x049\xFF
Backlight Mode	\x81\x09\x043\xFF
Power	\x81\x09\x04\x00\xFF

Appendix C. Set Commands (Ethernet)

Auto Focus Off	\x81\x01\x048\x03\xFF
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**Global Scripter Module
Communication Sheet**

Auto Focus On	\x81\x01\x048\x02\xff
Focus Far Speed 0	\x81\x01\x04\x08\xff
Focus Far Speed 7	\x81\x01\x04\x08'\xff
Focus Near Speed 0	\x81\x01\x04\x080\xff
Focus Near Speed 7	\x81\x01\x04\x087\xff
Focus Stop Speed 0	\x81\x01\x04\x08\x00\xff
Focus Stop Speed 7	\x81\x01\x04\x08\x00\xff
Pan-Tilt Down Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\xff
Pan-Tilt Down Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x01\x02\xff
Pan-Tilt Down Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x02\xff
Pan-Tilt Down Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x01\x02\xff
Pan-Tilt Down Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x02\xff
Pan-Tilt Down Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x02\xff
Pan-Tilt Down Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x02\xff
Pan-Tilt Down Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x02\xff
Pan-Tilt Down Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x02\xff
Pan-Tilt Down Right Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\x02\xff
Pan-Tilt Down Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x02\xff
Pan-Tilt Down Right Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\x02\xff
Pan-Tilt Home Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x04\xff
Pan-Tilt Home Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x04\xff
Pan-Tilt Home Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x04\xff
Pan-Tilt Home Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x04\xff
Pan-Tilt Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\xff
Pan-Tilt Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x01\x03\xff
Pan-Tilt Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x03\xff
Pan-Tilt Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x01\x03\xff
Pan-Tilt Stop Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x03\xff
Pan-Tilt Stop Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x03\xff
Pan-Tilt Stop Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x03\xff
Pan-Tilt Stop Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x03\xff
Pan-Tilt Up Left Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x01\xff
Pan-Tilt Up Left Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x01\x01\xff
Pan-Tilt Up Left Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x01\x01\xff
Pan-Tilt Up Left Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x01\x01\xff
Pan-Tilt Up Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x03\x01\xff
Pan-Tilt Up Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x03\x01\xff
Pan-Tilt Up Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x03\x01\xff
Pan-Tilt Up Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x03\x01\xff
Pan-Tilt Up Right Pan Speed 1 Tilt Speed 1	\x81\x01\x06\x01\x01\x01\x02\x01\xff
Pan-Tilt Up Right Pan Speed 1 Tilt Speed 20	\x81\x01\x06\x01\x01\x14\x02\x01\xff
Pan-Tilt Up Right Pan Speed 24 Tilt Speed 1	\x81\x01\x06\x01\x18\x01\x02\x01\xff
Pan-Tilt Up Right Pan Speed 24 Tilt Speed 20	\x81\x01\x06\x01\x18\x14\x02\x01\xff
Preset Recall Number 0	\x81\x01\x04?\x02\x00\xff
Preset Recall Number 127	\x81\x01\x04?\x02\x7F\xff
Preset Recall Number 254	\x81\x01\x04?\x02\xFE\xff
Preset Reset Number 0	\x81\x01\x04?\x00\x00\xff
Preset Reset Number 127	\x81\x01\x04?\x00\x7F\xff

Preset Reset Number 254	\x81\x01\x04?\x00\xFE\xFF
Preset Set Number 0	\x81\x01\x04?\x01\x00\xFF
Preset Set Number 127	\x81\x01\x04?\x01\x7F\xFF
Preset Set Number 254	\x81\x01\x04?\x01\xFE\xFF
Zoom Stop Speed 0	\x81\x01\x04\x07\x00\xFF
Zoom Stop Speed 7	\x81\x01\x04\x07\x00\xFF
Zoom Tele Speed 0	\x81\x01\x04\x07 \xFF
Zoom Tele Speed 7	\x81\x01\x04\x07'\xFF
Zoom Wide Speed 0	\x81\x01\x04\x070\xFF
Zoom Wide Speed 7	\x81\x01\x04\x077\xFF

Appendix D. Query Commands (Ethernet)

Required Polling	\x81\x09\x04G\xFF
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